Exam 2 Orders of the Day

Orthoptera!

Kristin Hilborn

Orthoptera

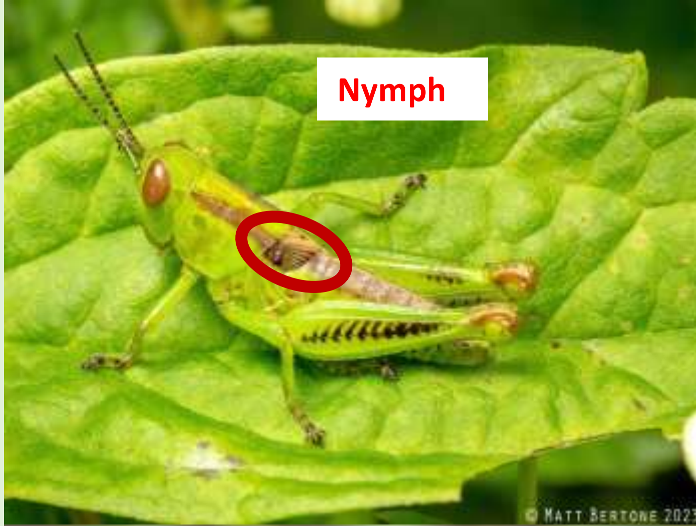
- “Straight-winged”
- ~ 300 million years old

Katydids & Crickets (Ensifera)

Grasshoppers and Pigmy Mole Crickets (Caelifera)

Hemimetabola

- Incomplete metamorphosis
- Egg → Nymph (several stages) → Adult
- Nymphs molt many times prior to adulthood
- Nymphs look very similar to adults; however, lack developed wings and reproductive organs
    - Developing wing pads are often visible in nymphs



Neoptera… the beginning of something new

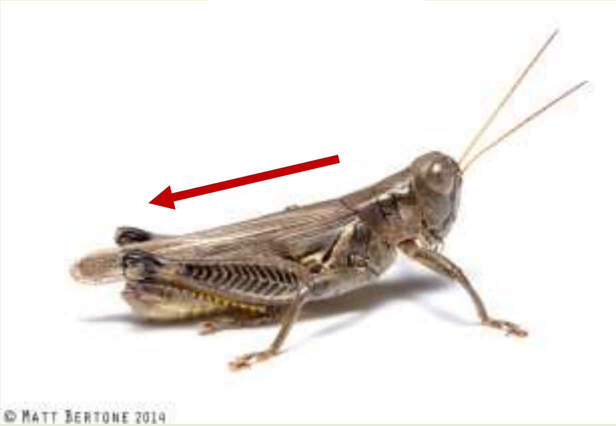
“Neo-” [New] = “Ptery”[Wing]

A new wing folding mechanism that allows a new resting position

- Prior to Orthopterans wings could only be held upright or outwards

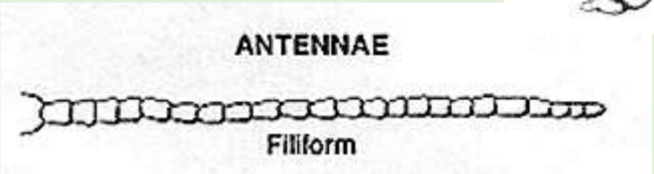
Before

After

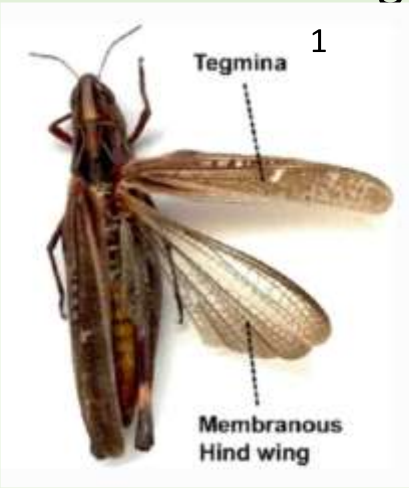


Orthoptera Characteristics

- Modified hind legs for jumping
- Mandibulate mouth parts
- Filiform antennae



- Forewing is leathery (tegmina) which covers membranous hind wing
- Ceri short and unsegmented
- Something called a Tympanum



[1]Chari, N., Rao, A.P., Srinivas, P., Girish, N. (2021). Types of Textures in Insect Wings and Classification. In: Chari, N., Mukkavilli, P., Parayitam, L. (eds) Biophysics of Insect Flight. Springer Series in Biophysics, vol 22. Springer, Singapore. https://doi.org/10.1007/978-981-16-5184-7_5

Orthoptera Characteristics

- Mostly Herbivorous
  - Some species “eat toxic plants, and keep the toxins in their bodies for protection”[2]
  - Few are carnivorous
- More than 25,000 species that are known today[2]
- Primarily solitary but some migratory species will gather in large numbers
- Most active during the day but still around at night
- Often lays eggs unground

[2]Web, A. D. (n.d.). *Critter Catalog*. BioKIDS. http://www.biokids.umich.edu/critters/Acrididae/

Fun Fact!

Orthopterans can be very large!

- The Giant Wētā can get up to 71g which is approx. 3x heavier than a mouse[2]
- Also, it's protected by law due to its risk of extinction

Dinobass, CC BY-SA 4.0 <https://creativecommons.org/licenses/by-sa/4.0>, via Wikimedia Commons

[2] *Top 10 largest insects in the world*. BBC Science Focus Magazine. (n.d.). https://www.sciencefocus.com/nature/largest-insects-in-the-

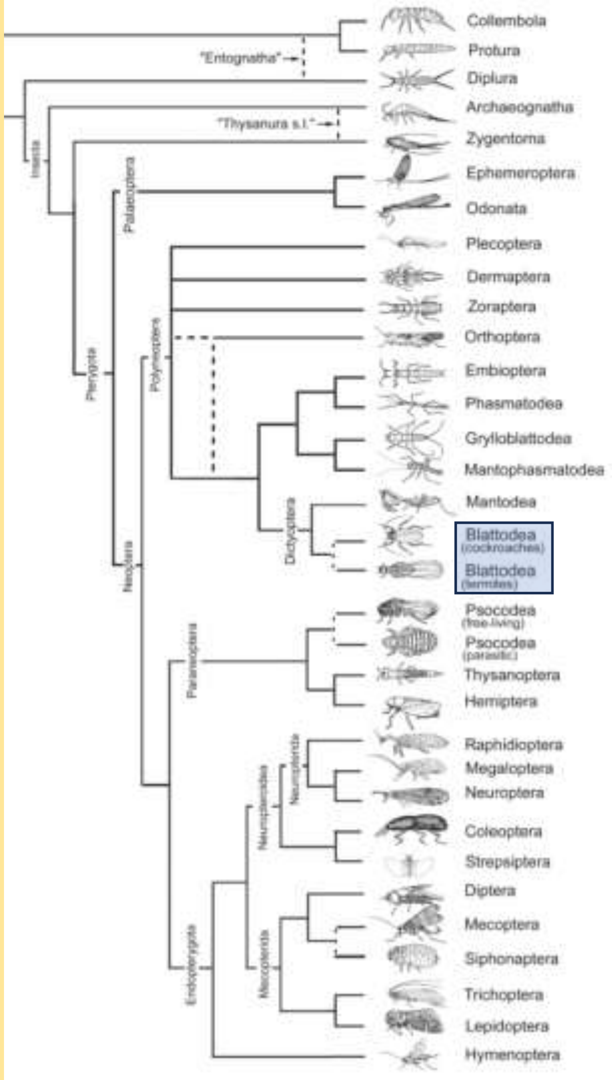
https://youtu.be/jWx1wPKAkpc?si=QubvHUJYjx7sPMWv

Blattodea & Isoptera

Alexis Alsdorf

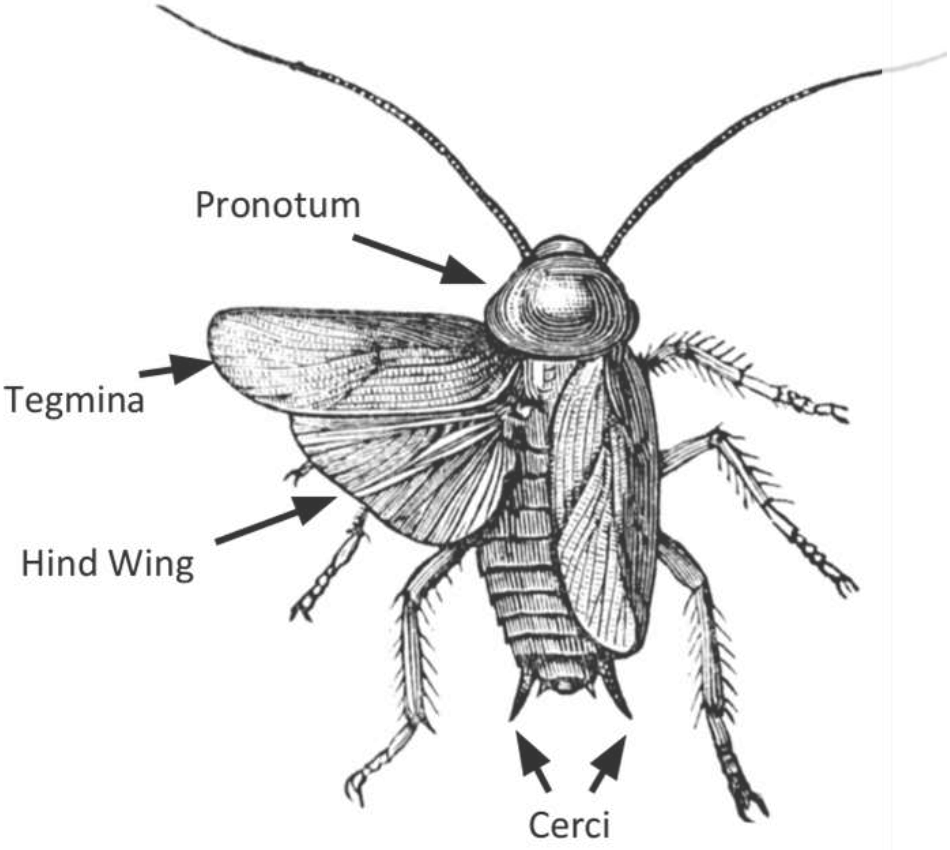
Blattodea – Cockroaches and Termites

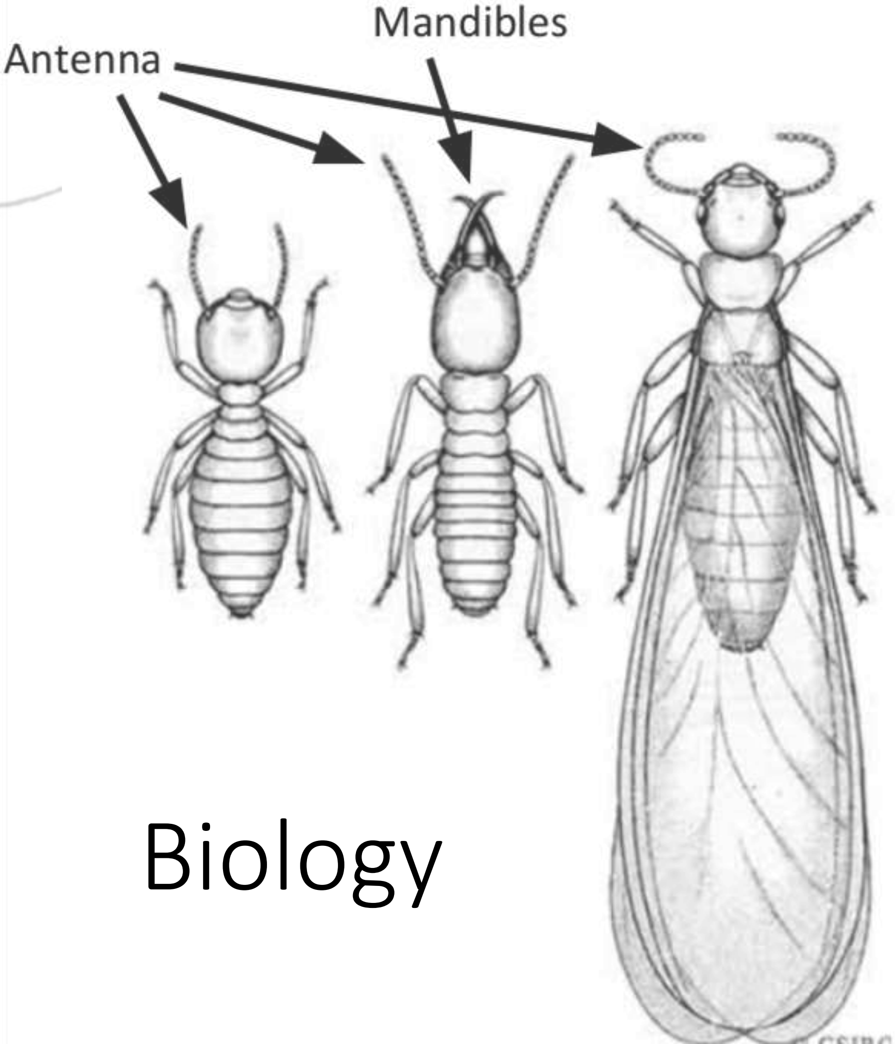
- “Blatta” – an insect that shuns the light
- Isoptera – 2007-2018
  - “iso” – equal
  - “ptera” - wings
- ~4,600 classified species of cockroach
- ~3,000 classified species of termites



Gullan & Cranston 2014

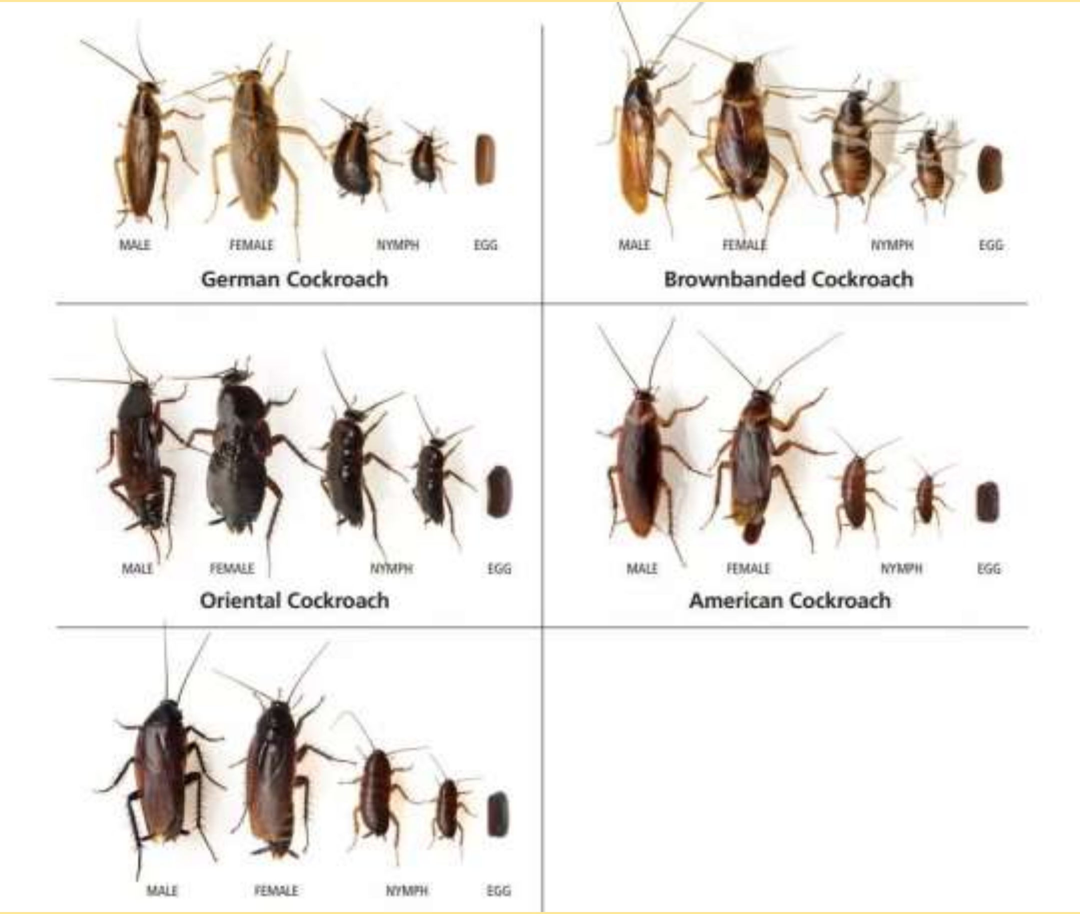
Biology





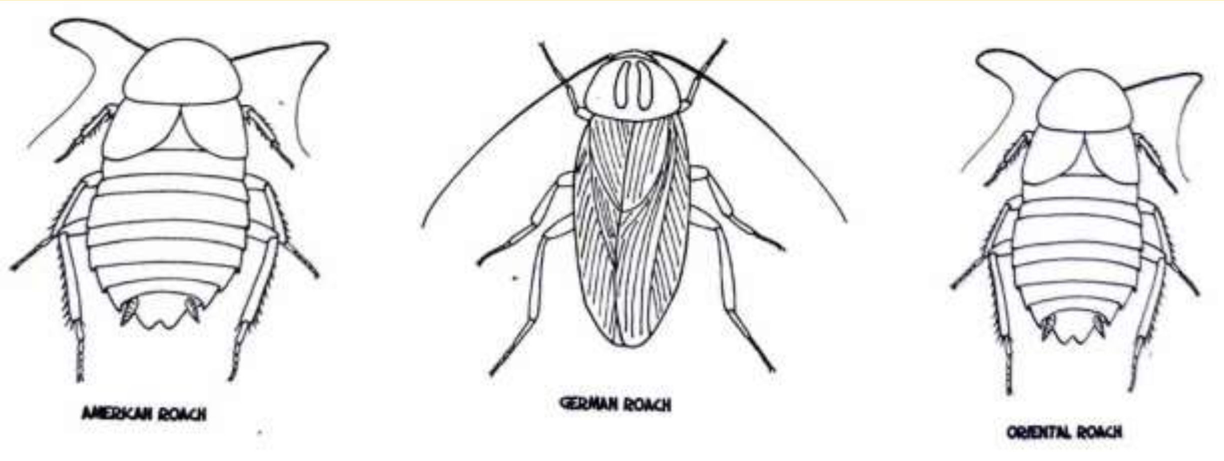
Cockroaches

- Hemimetabolous – incomplete metamorphosis
  - Eggs, nymphs, adults
- Ootheca
  - “oo” - egg
  - “theca” - receptacle
- Food
  - Leaf litter/Detritus
  - Wood
  - Last night’s dinner
  - Hair
  - Something with nutrition



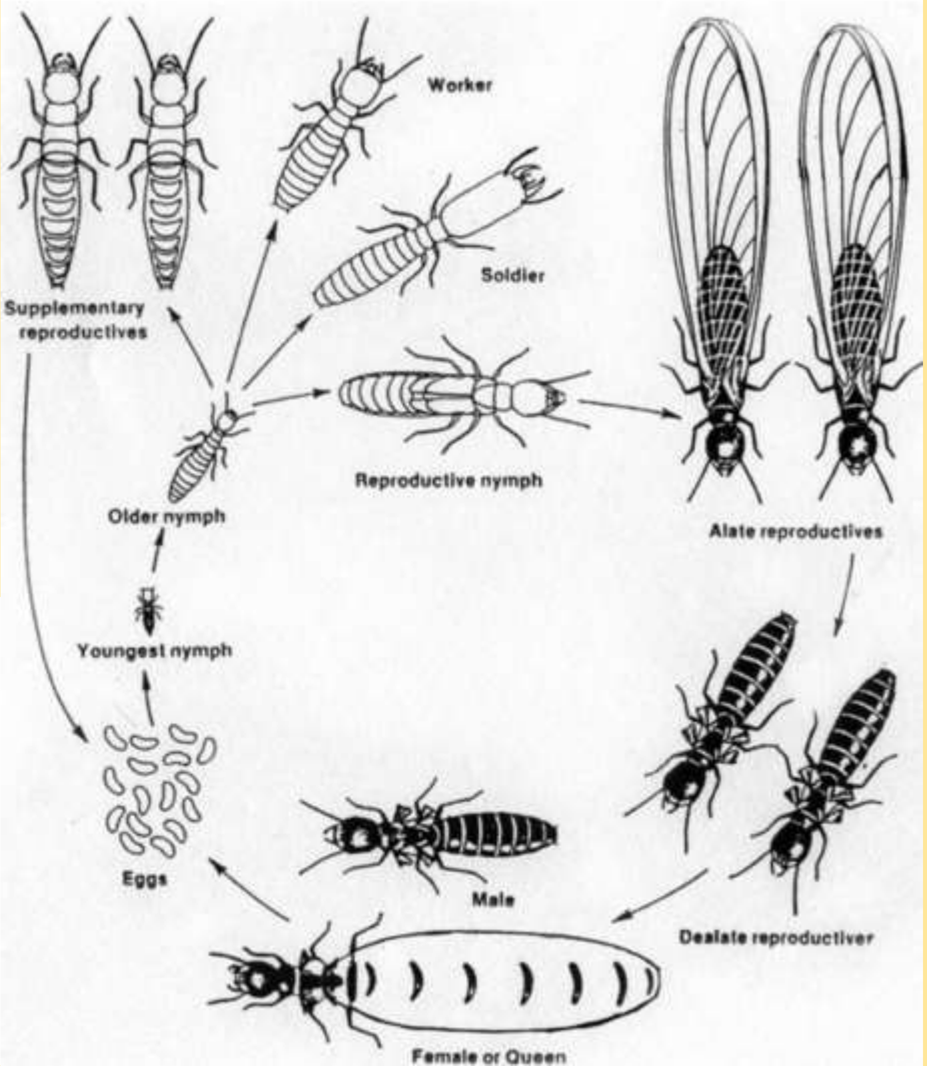
Cockroaches

- Interesting behaviors
  - Nuptial gifts
    - Tergal gland - produces a "treat" for the interested female
  - Kin recognition
  - Shared nest sites (aggregations) guided by pheromones
  - Some exhibit parental care
  - Fecal trail navigation
- Not all are pests



Termites (used to be Isoptera)

- Evolved ~170 – 54 mya
- Hemimetabolous
- “Eusocial Cockroaches”
  - Cooperative brood care
  - Division of labor
  - Overlapping generations
- “caste system”
  - Queen
  - King
  - Secondary Queen
  - Tertiary Queen
  - Soldiers
  - Workers



Termites

- Gut microbiome
  - Extremely diverse and dense
    - Breaks down wood
- Cellulose-based habitat

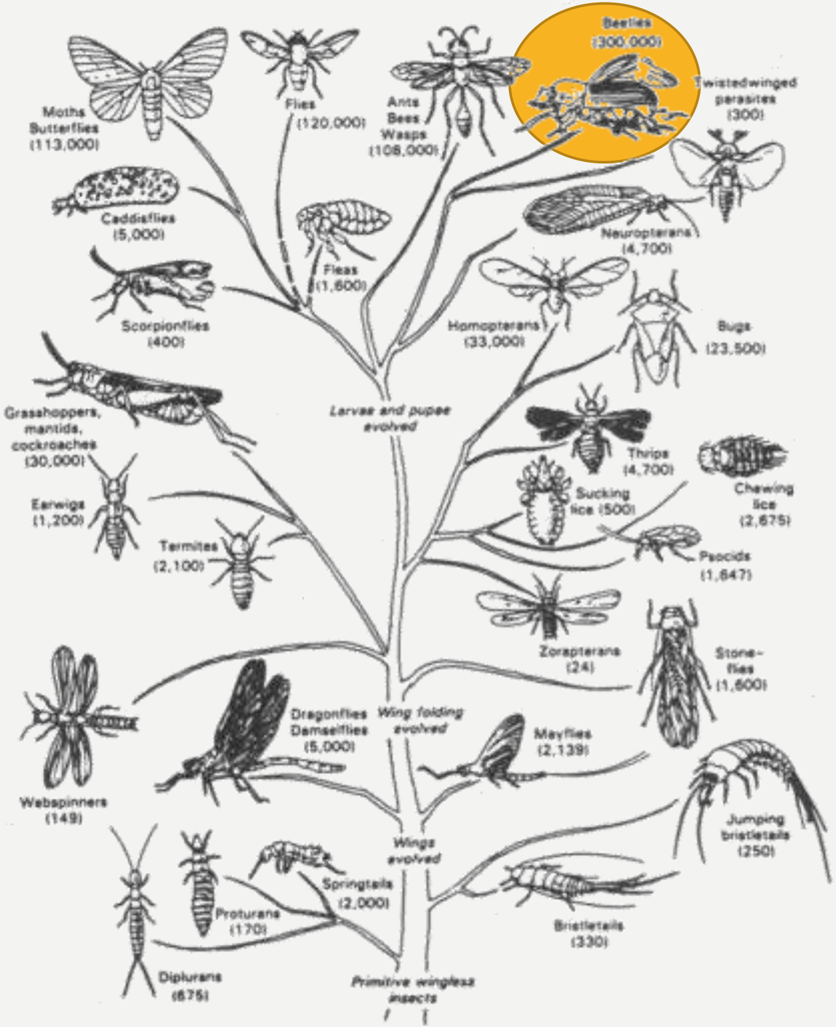
- https://www.youtube.com/watch?v=62OomdSZzBs

Ent 201

Order of the Day Presentation:

COLEOPTERA

TAYNARA POSSEBOM

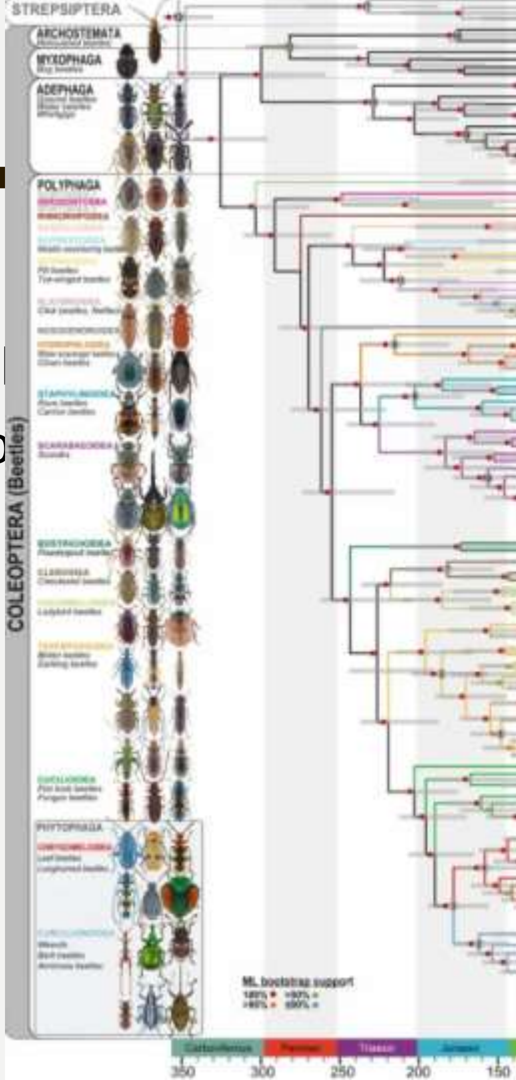


Beetles
(300,000)
Twistedwinged parasites
(300)
Moths
Butterflies
(113,000)
Flies
(120,000)
Ants
Bees
Wasps
(108,000)
Caddisflies
(5,000)
Fleas
(1,600)
Neuropterans
(4,700)
Scorpionflies
(400)
Homopterans
(33,000)
Bugs
(23,500)
Larvae and pupae
evolved
Grasshoppers,
mantids,
cockroaches
(30,000)
Thrips
(4,700)
Sucking
lice (500)
Chewing
lice
(2,675)
Earwigs
(1,200)
Termites
(2,100)
Psocids
(1,647)
Zorapterans
(24)
Stone-
flies
(1,600)
Dragonflies
Damselflies
(5,000)
Wing folding
evolved
Mayflies
(2,139)
Webspinners
(149)
Jumping
bristletails
(250)
Wings
evolved
Springtails
(2,000)
Proturans
(170)
Bristletails
(330)
Diplurans
(675)
Primitive wingless
insects

OH, TO KNOW A BEETL

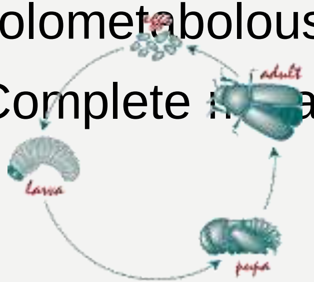
- Originated from Greek → "koleos" – *sheath* and "
- Stepped onto the scene ~285mya (Permian perio
- Largest insect order (description bias)
  - ~400,000 described species
  - Next la with ~180,000 species

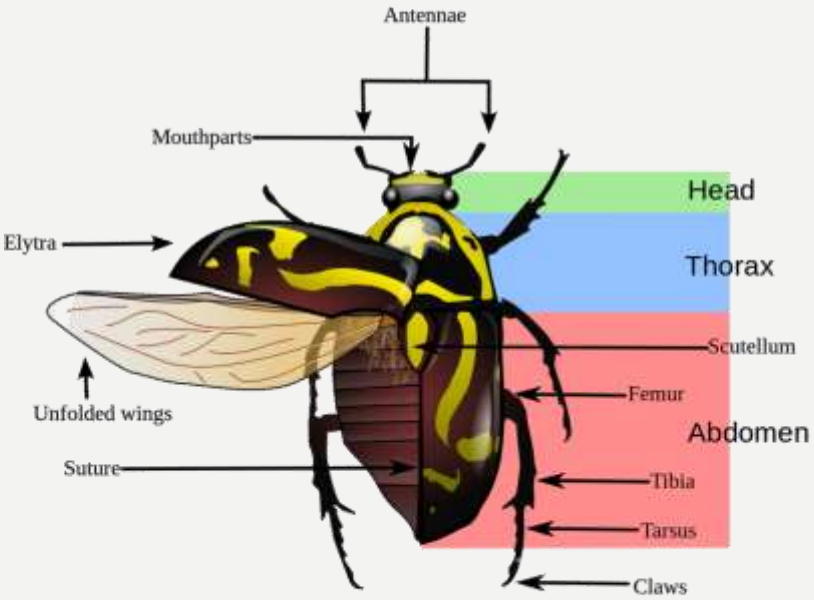
Oldest crown-group fossil ~230 Ma old



THE BEETLE BODY & DEVELOPMENT

- Characterized by their hardened forewing → The <u>Elytra</u>
  - Their delicate hind wings are tightly folded under the elytra, waiting for flight!
- <u>Scutellum</u> is the top plate (tergite) of their mesothorax
- Holometabolous
- Complete metamorphosis





THE "SUPER" FAMILIES

- The order Coleoptera is subdivided into many super families
  - Often these groups are INCREDIBLY large, and can be HIGHLY diverse

- Examples:

- Curculionidae (Weevils) → ~97,000 species
- Staphylinidae (Rove Beetles) → ~67,000 speci
- Tenebrionidae (Darkling Beetles) → ~20,000 sp

Curculio occidentis (Oak Weevil)

THE "SUPER" FAMILIES

- The order Coleoptera is subdivided into many super families
  - Often these groups are INCREDIBLY large, and can be HIGHLY diverse

- Examples:

- Curculionidae (Weevils) → ~97,000 species
- Staphylinidae (Rove Beetles) → ~67,000 species
- Tenebrionidae (Darkling Beetles) → ~20,000 species

Paederus spp. (Highly inflammatory hemolymph)

THE "SUPER" FAMILIES

- The order Coleoptera is subdivided into many super families
  - Often these groups are INCREDIBLY large, and can be HIGHLY diverse

- Examples:

- Curculionidae (Weevils) → ~97,000 species
- Staphylinidae (Rove Beetles) → ~67,000 specie
- Tenebrionidae (Darkling Beetles) → ~20,000 sp

Asbolus verrucosus (B-Death Feigning Beetle)

A lot more!

MASTERS OF DIVERSITY

- Diverse Morphologies!
  - Antenna (*Rhipiceridae* – Cedar Beetles)
  - Legs (Harlequin Beetle)
  - Pronotum/Thorax (Rhinoceros/Stag Beetle; Giraffe Weevil)

MASTERS OF DIVERSITY

- Diverse Morphologies!

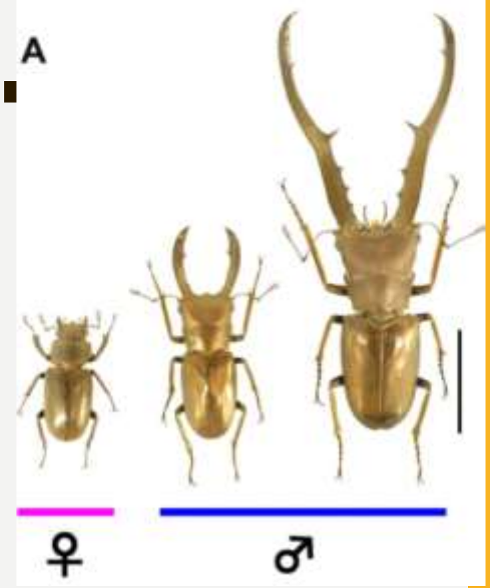
Antenna (*Rhipiceridae* – Cedar

MASTERS OF DIVERSITY

- Diverse Morphologies
  - Legs (Harlequin B

MASTERS OF DIVERSI

- Diverse Morphologies!
  - Pronotum/Thorax (Rhinoceros/Stag Beetle; Giraffe



MASTERS OF DIVERSITY

Diverse Habitats/Food

- Carnivores
- Plant feeders
- Parasitoids (Rhipiphori
- Dung/Detritus

MASTERS OF DIVERSITY

Diverse Habitats/Food

- Carnivores
- Plant feeders
- Parasitoids (Rhipiphori
- Dung/Detritus

MASTERS OF DIVERSITY

Diverse Habitats/Food

- Carnivores
- Plant feeders
- Parasitoids (Ripiphori
- Dung/Detritus

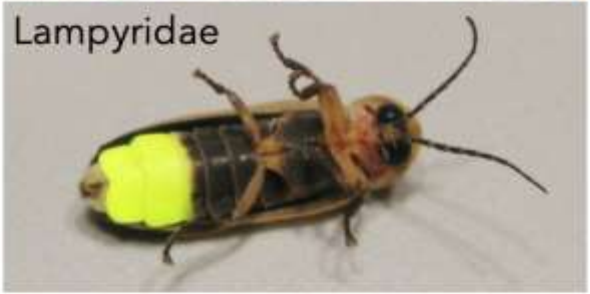
MASTERS OF DIVERSITY

Diverse Habitats/Food

- Carnivores
- Plant feeders
- Parasitoids (Rhipiph
- Dung/Detritus

MASTERS OF DIVERS

- Three beetle families exhibit bioluminescence



BEETLES IN YOUR BACKYARD

- Many species overlap with humans in both positive and negative ways, some you already know!

Ladybug → Coccinellidae

An excellent beetle to have present in gardens due to it is

Dung Beetle → Scarabs

Responsible for keeping our world from piling up with

CPB → Chrysomelidae

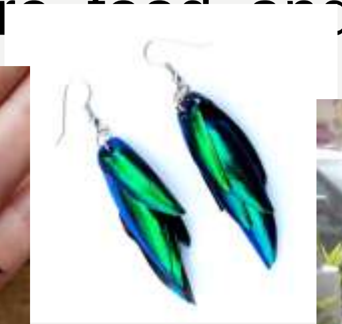
An incredibly tenacious and <u>resistant</u> pest

Boll Weevil→ Curculionidae

Major pest of cotton, whose eradication was achieved through an incredible effort!

THE COOL BEETLES!

- The order Coleoptera has also had a major impact on society, pop culture, food, and technology!



THE COOL BEETLES!

- The order Coleoptera has also had a major impact on society, pop culture, food, and technology!

THE COOL BEETLES!

- The order Coleoptera has also had a major impact on society, pop culture, food, and technology!


BIOTECHNOLOGY

The Army's Remote-Controlled Beetle

The insect's flight path can be wirelessly controlled via a neural implant.

By Emily Singer

January 29, 2009

IS THIS AFRICAN INSECT THE STRONGEST IN THE WORLD?

https://www.youtube.com/watch?v=4hXe3DLlS3k

THE GREAT DIVING BEETLE UNDERWATER HUNTER!

https://www.youtube.com/watch?v=Hwqja6FWUzA

QUESTIONS?